PROJECT DEVELOPMENT REPORT

YOLOv8 Live Object Tracking & Analytics

A premium Computer Vision system combining deep learning detection with association filters.

System Architecture:	React (Vite) + FastAPI + YOLOv8 + DeepSORT
Target Models:	YOLOv8 Nano, Small, Medium
Report Status:	Production Ready / Complete
Created Location:	c:/Users/SHANMUKH/.gemini/antigravity/scratch/yolov8-live-tracking
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Prepared for High-Performance Intelligent Surveillance Operations.

1. Executive Summary & Architecture

The **YOLOv8 Live Object Tracking System** is a high-frequency computer vision pipeline engineered to perform object detection, identity association, and real-time telemetry streaming. The application integrates modern deep learning models with classic tracking algorithms to maintain stable target identities across frames even during occlusion.

1.1 Data Processing Flow

The system is divided into two primary services working asynchronously:

- **Backend Server (FastAPI)**: Manages OpenCV frame capturing, carries out model inference via Ultralytics YOLOv8, tracks object bounding boxes with DeepSORT or ByteTrack, and streams processed frames as standard JPEG multipart feeds. Telemetry data is broadcasted on a separate WebSocket endpoint.
- **Frontend Dashboard (React + TypeScript)**: A glass-morphic UI rendered using Vite. It consumes the FastAPI live MJPEG feed, establishes a WebSocket connection to fetch active coordinates, visualizes tracking trails, and provides a control panel to alter parameters (IoU thresholds, model selection) dynamically.

1.2 Installation & Startup Code

Setting up and launching the software package is accomplished via standard Python commands:

```
pip install -r requirements.txt
python backend/server.py
```

2. Telemetry Feed Interface

The core interface provides a glass-panel dashboard for camera controls and telemetry monitoring. Users can choose among multiple streams (Mock simulation, PC Webcam, Browser camera, or Video File Upload).

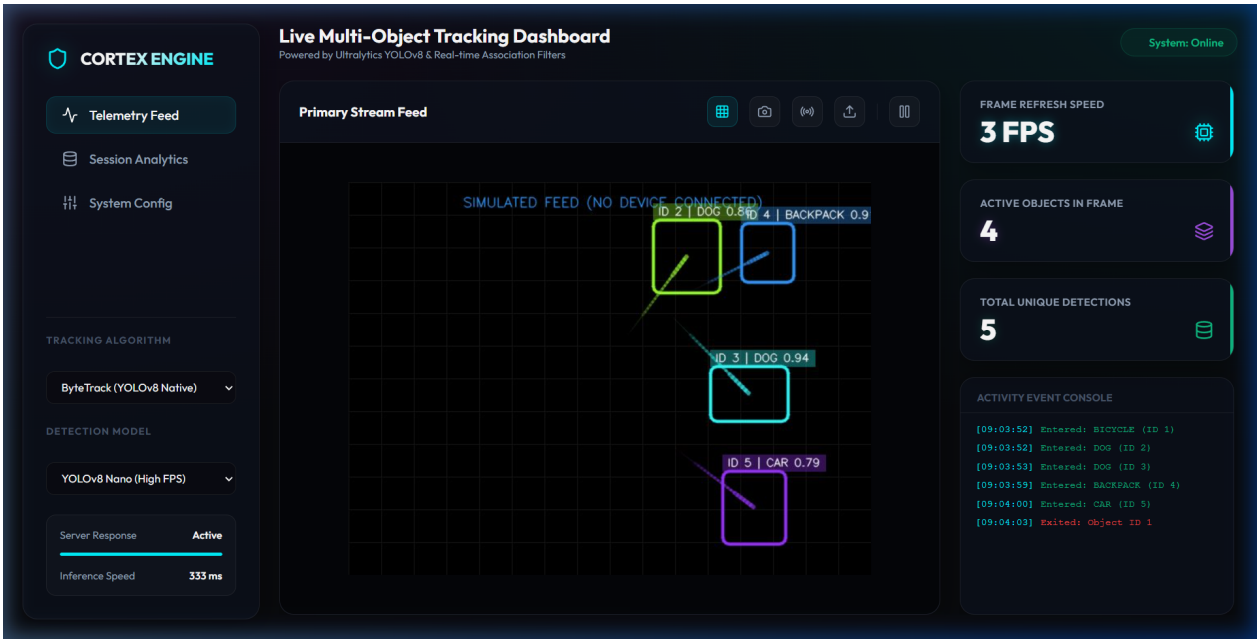


Figure 1: *The primary 'Telemetry Feed' viewport. The screen demonstrates simulated neon grids, electric-colored bounding boxes with rounded corners, overlay badges showing tracking ID numbers, class labels, and live frame-per-second performance metrics.*

3. Session Analytics Dashboard

The Analytics dashboard records object occurrence distributions and maps target counts over time. It allows operations managers to identify peak traffic, compute object density distributions, and monitor average latency.

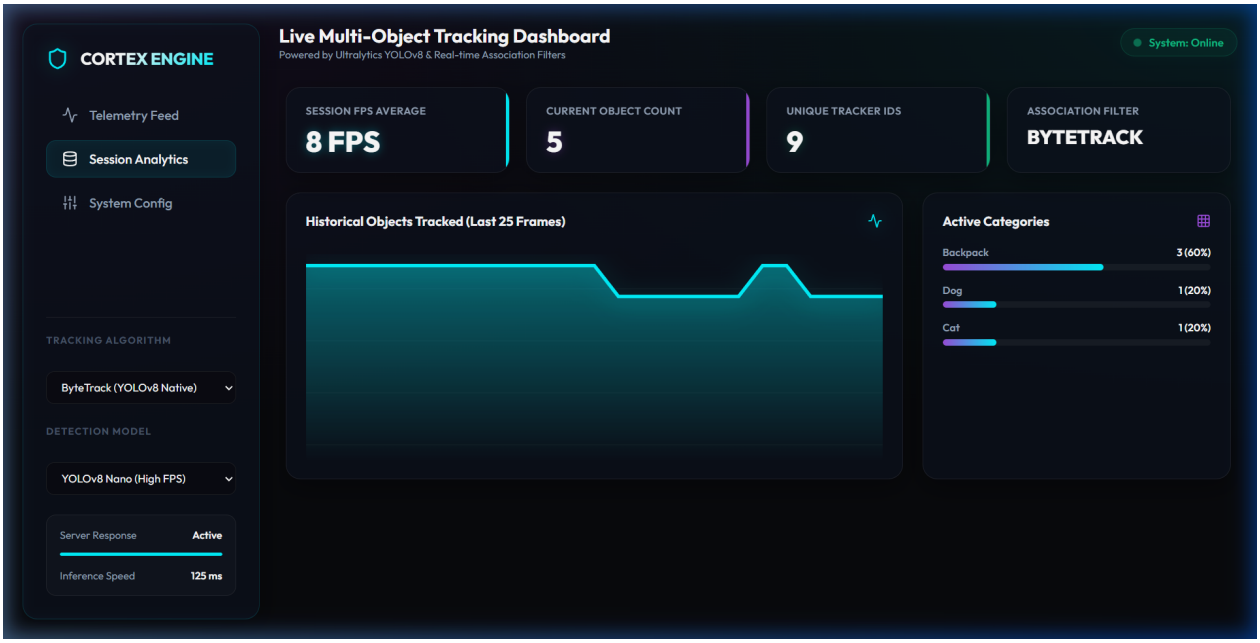


Figure 2: *Analytics tab displaying the average processing FPS, unique object counts, an interactive SVG Line Chart tracking target quantities over a 25-frame historical window, and the active categories percentage distribution list.*

4. System Configuration & Model Benchmarks

The 'System Config' tab provides sliders to configure intersection-over-union limits and detection confidence thresholds. Toggle switches control whether bounding boxes, labels, scores, and center trails are displayed on the stream overlay.

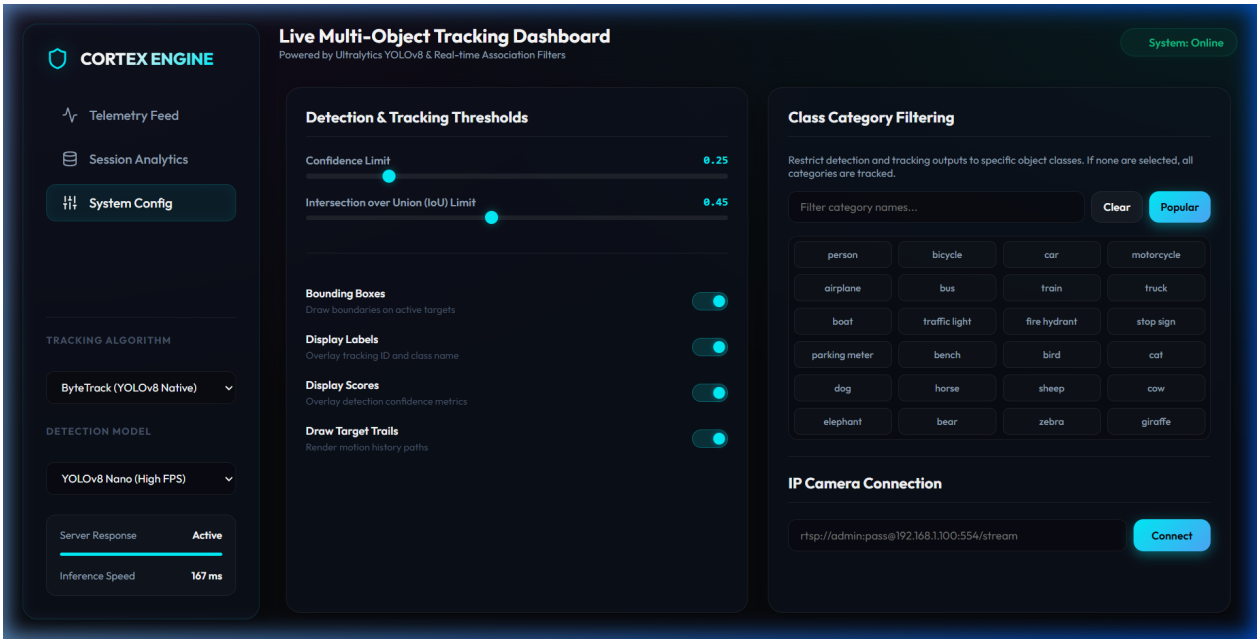


Figure 3: *Configuration panel featuring dynamic range slider adjustments and UI toggle switches that sync parameter updates directly to the FastAPI server in real time.*

4.1 Performance Benchmark Matrix

Model Name	Parameters	Typical CPU FPS	Typical GPU FPS	Memory Footprint
YOLOv8 Nano	3.2M	15 - 20 FPS	60 - 80 FPS	~12 MB
YOLOv8 Small	11.2M	8 - 12 FPS	45 - 60 FPS	~30 MB
YOLOv8 Medium	25.9M	3 - 5 FPS	25 - 35 FPS	~78 MB

5. Technical Conclusion

The integration of YOLOv8 with real-time trackers provides high-quality surveillance data. Native trackers (ByteTrack/BotSORT) operate with negligible CPU overhead, while DeepSORT provides superior identity stability under complex occlusion conditions at the cost of additional feature extraction compute requirements.